

▼ A. Olympics

```
!pip install -q pyspark
```

```
from pyspark.sql import SparkSession

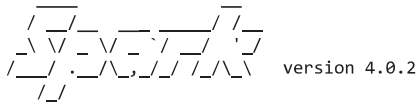
spark = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()

print("Spark Version:", spark.version)
```

Spark Version: 4.0.2

```
!pyspark --version
```

WARNING: Using incubator modules: jdk.incubator.vector
Welcome to



Using Scala version 2.13.16, OpenJDK 64-Bit Server VM, 17.0.18
Branch HEAD
Compiled by user runner on 2026-02-02T08:08:13Z
Revision 7cc3b9bcdab8c923f23cdbc9ce922530e1becf1
Url <https://github.com/apache/spark>
Type --help for more information.

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
olympic = spark.read.csv("/content/drive/MyDrive/Datasets/olympix_data.csv", sep = "\t", header = True, inferSchema=True)
```

```
olympic.show()
```

Athlete	Age	Country	Year	Closing Ceremony Date	Sport	Gold Medals	Silver Medals	Bronze Medals
Michael Phelps	23	United States	2008	08-24-08	Swimming	8	0	0
Michael Phelps	19	United States	2004	08-29-04	Swimming	6	0	2
Michael Phelps	27	United States	2012	08-12-12	Swimming	4	2	0
Natalie Coughlin	25	United States	2008	08-24-08	Swimming	1	2	3
Aleksey Nemov	24	Russia	2000	10-01-00	Gymnastics	2	1	3
Alicia Coutts	24	Australia	2012	08-12-12	Swimming	1	3	1
Missy Franklin	17	United States	2012	08-12-12	Swimming	4	0	1
Ryan Lochte	27	United States	2012	08-12-12	Swimming	2	2	1
Allison Schmitt	22	United States	2012	08-12-12	Swimming	3	1	1
Natalie Coughlin	21	United States	2004	08-29-04	Swimming	2	2	1
Ian Thorpe	17	Australia	2000	10-01-00	Swimming	3	2	0
Dara Torres	33	United States	2000	10-01-00	Swimming	2	0	3
Cindy Klassen	26	Canada	2006	02-26-06	Speed Skating	1	2	2
Nastia Liukin	18	United States	2008	08-24-08	Gymnastics	1	3	1
Marit Bjørgen	29	Norway	2010	02-28-10	Cross Country Skiing	3	1	1
Sun Yang	20	China	2012	08-12-12	Swimming	2	1	1
Kirsty Coventry	24	Zimbabwe	2008	08-24-08	Swimming	1	3	0
Libby Lenton-Tric...	23	Australia	2008	08-24-08	Swimming	2	1	1
Ryan Lochte	24	United States	2008	08-24-08	Swimming	2	0	2
Inge de Bruijn	30	Netherlands	2004	08-29-04	Swimming	1	1	2

only showing top 20 rows

```
olympic.printSchema()
```

```
root
|-- Athlete: string (nullable = true)
|-- Age: integer (nullable = true)
|-- Country: string (nullable = true)
```

```
-- Year: integer (nullable = true)
-- Closing Ceremony Date: string (nullable = true)
-- Sport: string (nullable = true)
-- Gold Medals: integer (nullable = true)
-- Silver Medals: integer (nullable = true)
-- Bronze Medals: integer (nullable = true)
-- Total Medals: integer (nullable = true)
```

```
olympic.show(5)
```

	Athlete	Age	Country	Year	Closing Ceremony Date	Sport	Gold Medals	Silver Medals	Bronze Medals	Total Medals
	Michael Phelps	23	United States	2008	08-24-08	Swimming	8	0	0	8
	Michael Phelps	19	United States	2004	08-29-04	Swimming	6	0	2	8
	Michael Phelps	27	United States	2012	08-12-12	Swimming	4	2	0	6
	Natalie Coughlin	25	United States	2008	08-24-08	Swimming	1	2	3	6
	Aleksey Nemov	24	Russia	2000	10-01-00	Gymnastics	2	1	3	6

```
only showing top 5 rows
```

```
olympic.select('Athlete', 'Country', 'Sport').show()
```

	Athlete	Country	Sport
	Michael Phelps	United States	Swimming
	Michael Phelps	United States	Swimming
	Michael Phelps	United States	Swimming
	Natalie Coughlin	United States	Swimming
	Aleksey Nemov	Russia	Gymnastics
	Alicia Coutts	Australia	Swimming
	Missy Franklin	United States	Swimming
	Ryan Lochte	United States	Swimming
	Allison Schmitt	United States	Swimming
	Natalie Coughlin	United States	Swimming
	Ian Thorpe	Australia	Swimming
	Dara Torres	United States	Swimming
	Cindy Klassen	Canada	Speed Skating
	Nastia Liukin	United States	Gymnastics
	Marit Bjørgen	Norway	Cross Country Skiing
	Sun Yang	China	Swimming
	Kirsty Coventry	Zimbabwe	Swimming
	Libby Lenton-Tric...	Australia	Swimming
	Ryan Lochte	United States	Swimming
	Inge de Bruijn	Netherlands	Swimming

```
only showing top 20 rows
```

```
olympic.createOrReplaceTempView("olympic_data")
```

```
spark.sql("SELECT Athlete, `Total Medals` FROM olympic_data WHERE `Total Medals` > 5").show()
```

	Athlete	Total Medals
	Michael Phelps	8
	Michael Phelps	8
	Michael Phelps	6
	Natalie Coughlin	6
	Aleksey Nemov	6

```
from pyspark.sql.functions import sum
```

```
country_medals = olympic.groupBy('Country').agg(sum('Total Medals').alias('TotalMedals'))
country_medals.orderBy('TotalMedals', ascending=False).show()
```

	Country	TotalMedals
	United States	1312
	Russia	768
	Germany	629
	Australia	609
	China	530
	Canada	370
	Italy	331

Great Britain	322
France	318
Netherlands	318
South Korea	308
Japan	282
Brazil	221
Spain	205
Norway	192
Cuba	188
Sweden	181
Hungary	145
Ukraine	143
Argentina	141

+-----+

only showing top 20 rows

```
from pyspark.sql.functions import avg

average_age_per_sport = olympic.groupBy('Sport').agg(avg('Age').alias('AverageAge'))
average_age_per_sport.orderBy('AverageAge', ascending=False).show()
```

Sport	AverageAge
Equestrian	37.92356687898089
Curling	32.853658536585364
Shooting	31.066298342541437
Sailing	29.952380952380953
Beach Volleyball	29.854166666666668
Bobsleigh	29.666666666666668
Skeleton	29.611111111111111
Luge	28.833333333333332
Biathlon	28.819148936170212
Cross Country Skiing	28.390625
Handball	28.225071225071225
Triathlon	28.125
Rowing	27.777777777777778
Cycling	27.367816091954023
Volleyball	27.269784172661872
Modern Pentathlon	27.25
Basketball	27.05574912891986
Baseball	27.050925925925927
Softball	27.044776119402986
Ice Hockey	26.9296875

+-----+

only showing top 20 rows

Start coding or [generate](#) with AI.

```
from pyspark.sql.functions import sum

# Group by Athlete and sum the Gold Medals, then order by the sum in descending order
athlete_gold_medals = olympic.groupBy('Athlete').agg(sum('Gold Medals').alias('TotalGoldMedals'))
athlete_gold_medals.orderBy('TotalGoldMedals', ascending=False).show(1)
```

Athlete	TotalGoldMedals
Michael Phelps	18

+-----+

only showing top 1 row

```
from pyspark.sql.functions import sum

# Group by Athlete and sum the Total Medals, then order by the sum in descending order
athlete_total_medals = olympic.groupBy('Athlete').agg(sum('Total Medals').alias('TotalMedals'))
athlete_total_medals.orderBy('TotalMedals', ascending=False).show()
```

Athlete	TotalMedals
Michael Phelps	22
Natalie Coughlin	12
Ryan Lochte	11
Ian Thorpe	9
Leisel Jones	9
Ole Einar Bjørndalen	9
Apolo Anton Ohno	8
Katalin Kovács	8
Dara Torres	8

Jason Lezak	8
Inge de Bruijn	8
Libby Lenton-Tric...	7
Kirsty Coventry	7
Veronica Campbell...	7
Felix Gottwald	7
Petria Thomas	7
Kosuke Kitajima	7
Yang Wei	7
Valentina Vezzali	7
Grant Hackett	7

only showing top 20 rows

```
from pyspark.sql.functions import countDistinct
```

```
# Group by Country and count the distinct Athletes
```

```
athletes_per_country = olympic.groupBy('Country').agg(countDistinct('Athlete').alias('NumberOfAthletes'))
```

```
athletes_per_country.orderBy('NumberOfAthletes', ascending=False).show()
```

Country	NumberOfAthletes
United States	878
Russia	567
Germany	422
Australia	415
China	358
Canada	282
Italy	247
France	239
Great Britain	232
South Korea	232
Japan	219
Netherlands	215
Spain	173
Brazil	165
Cuba	153
Sweden	143
Norway	127
Ukraine	124
Argentina	103
Finland	92

only showing top 20 rows

```
from pyspark.sql.functions import col
```

```
# Filter for athletes who won at least one of each medal type
```

```
at_least_one_of_each = olympic.filter(
    (col('Gold Medals') >= 1) &
    (col('Silver Medals') >= 1) &
    (col('Bronze Medals') >= 1)
)
```

```
# Show distinct athletes who meet the criteria
```

```
at_least_one_of_each.select('Athlete').distinct().show()
```

Athlete
Libby Lenton-Tric...
Georg Hettich
Aliya Mustafina
Roland Schoeman
Ryan Lochte
Laure Manaudou
Kirsty Coventry
Kaitlin Sandeno
Ian Thorpe
Yuliya Chepalova
Allison Schmitt
Bode Miller
Leisel Jones
Gary Hall Jr.
Alain Bernard
Dmitry Sautin
Wang Meng
Ian Crocker
Chad Hedrick
Massimiliano Roso...

+-----+
only showing top 20 rows

▼ Players

```
players = spark.read.csv("/content/drive/MyDrive/Datasets/players (1).csv", sep = ",", header = True, inferSchema=True)
```

```
players.show()
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
3	Neymar Jr	NULL	Brazil	15	Al Hilal
4	Kylian Mbappe	25	NULL	30	PSG
5	Kevin De Bruyne	32	Belgium	NULL	Man City
6	NULL	29	France	18	PSG
7	Luka Modric	38	Croatia	10	NULL
8	NULL	NULL	NULL	NULL	NULL
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
11	Karim Benzema	36	France	NULL	Al Ittihad
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
14	Eden Hazard	33	Belgium	NULL	Real Madrid
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
20	Gavi	19	Spain	NULL	Barcelona

only showing top 20 rows

```
players.printSchema()
```

```
root
|-- PlayerID: integer (nullable = true)
|-- Name: string (nullable = true)
|-- Age: integer (nullable = true)
|-- Country: string (nullable = true)
|-- Goals: integer (nullable = true)
|-- Club: string (nullable = true)
```

Start coding or [generate](#) with AI.

```
from pyspark.sql.functions import col, when, count

# Count null values for each column
missing_values = players.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in players.columns
])

missing_values.show()
```

PlayerID	Name	Age	Country	Goals	Club
0	2	2	2	6	3

```
print(f"Number of rows before dropping nulls: {players.count()}")

players_cleaned = players.dropna()

print(f"Number of rows after dropping nulls: {players_cleaned.count()}")
players_cleaned.show()
```

```
Number of rows before dropping nulls: 30
Number of rows after dropping nulls: 19
+-----+
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
21	Son Heung-min	32	South Korea	16	Tottenham
22	Bruno Fernandes	30	Portugal	9	Man United
23	Marcus Rashford	27	England	13	Man United
24	Jude Bellingham	21	England	17	Real Madrid
25	Phil Foden	24	England	15	Man City
26	Joao Felix	26	Portugal	6	Atletico Madrid
27	Antoine Griezmann	34	France	12	Atletico Madrid
28	Olivier Giroud	38	France	10	AC Milan

```
print(f"Number of rows before dropping rows with all nulls: {players.count()}")
```

```
players_cleaned_all_nulls = players.dropna(how='all')
```

```
print(f"Number of rows after dropping rows with all nulls: {players_cleaned_all_nulls.count()}")
players_cleaned_all_nulls.show()
```

```
Number of rows before dropping rows with all nulls: 30
Number of rows after dropping rows with all nulls: 30
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami
2	Cristiano Ronaldo	38	Portugal	25	Al Nassr
3	Neymar Jr	NULL	Brazil	15	Al Hilal
4	Kylian Mbappe	25	NULL	30	PSG
5	Kevin De Bruyne	32	Belgium	NULL	Man City
6	NULL	29	France	18	PSG
7	Luka Modric	38	Croatia	10	NULL
8	NULL	NULL	NULL	NULL	NULL
9	Erling Haaland	23	Norway	28	Man City
10	Harry Kane	30	England	24	Bayern Munich
11	Karim Benzema	36	France	NULL	Al Ittihad
12	Paulo Dybala	30	Argentina	12	Roma
13	Thomas Muller	34	Germany	8	Bayern Munich
14	Eden Hazard	33	Belgium	NULL	Real Madrid
15	Gerard Moreno	31	Spain	11	Villarreal
16	Sadio Mane	32	Senegal	14	Al Nassr
17	Mohamed Salah	32	Egypt	21	Liverpool
18	Virgil van Dijk	33	Netherlands	5	Liverpool
19	Pedri	21	Spain	7	Barcelona
20	Gavi	19	Spain	NULL	Barcelona

```
only showing top 20 rows
```

```
players_filled = players.fillna({"Goals": 0, "Club": "Unknown"})
```

```
print("Schema after filling missing values:")
players_filled.printSchema()
```

```
print("\nDataFrame after filling missing values (first 20 rows):")
players_filled.show()
```

```
Schema after filling missing values:
```

```
root
```

```
-- PlayerID: integer (nullable = true)
-- Name: string (nullable = true)
-- Age: integer (nullable = true)
-- Country: string (nullable = true)
-- Goals: integer (nullable = false)
-- Club: string (nullable = false)
```

```
DataFrame after filling missing values (first 20 rows):
```

PlayerID	Name	Age	Country	Goals	Club
1	Lionel Messi	36	Argentina	20	Inter Miami

```

|      2|Cristiano Ronaldo| 38| Portugal| 25| Al Nassr|
|      3|      Neymar Jr|NULL|  Brazil| 15| Al Hilal|
|      4|    Kylian Mbappe| 25|    NULL| 30|    PSG|
|      5|    Kevin De Bruyne| 32| Belgium|  0| Man City|
|      6|      NULL| 29|  France| 18|    PSG|
|      7|    Luka Modric| 38| Croatia| 10| Unknown|
|      8|      NULL|NULL|    NULL|  0| Unknown|
|      9|    Erling Haaland| 23| Norway| 28| Man City|
|     10|    Harry Kane| 30| England| 24|Bayern Munich|
|     11|    Karim Benzema| 36|  France|  0| Al Ittihad|
|     12|    Paulo Dybala| 30| Argentina| 12|    Roma|
|     13|    Thomas Muller| 34| Germany|  8|Bayern Munich|
|     14|    Eden Hazard| 33| Belgium|  0| Real Madrid|
|     15|    Gerard Moreno| 31| Spain| 11| Villarreal|
|     16|    Sadio Mane| 32| Senegal| 14| Al Nassr|
|     17|    Mohamed Salah| 32| Egypt| 21| Liverpool|
|     18|    Virgil van Dijk| 33|Netherlands| 5| Liverpool|
|     19|      Pedri| 21| Spain| 7| Barcelona|
|     20|      Gavi| 19| Spain|  0| Barcelona|
+-----+-----+-----+-----+-----+
only showing top 20 rows

```

Now, let's confirm the missing values after the fill operation for 'Goals' and 'Club'.

```

from pyspark.sql.functions import col, when, count

# Re-count null values to confirm fillna operation
missing_values_after_fill = players_filled.select([
    count.when(col(c).isNull(), c).alias(c)
    for c in players_filled.columns
])

print("\nMissing values after fillna:")
missing_values_after_fill.show()

```

```

Missing values after fillna:
+-----+-----+-----+-----+-----+
|PlayerID|Name|Age|Country|Goals|Club|
+-----+-----+-----+-----+-----+
|      0|  2|  2|    2|  0|  0|
+-----+-----+-----+-----+-----+

```

```

players_with_more_than_20_goals = players_filled.filter(col('Goals') > 20)
players_with_more_than_20_goals.show()

```

```

+-----+-----+-----+-----+-----+
|PlayerID|      Name|Age| Country|Goals|      Club|
+-----+-----+-----+-----+-----+
|      2|Cristiano Ronaldo| 38|Portugal|  25|    Al Nassr|
|      4|    Kylian Mbappe| 25|    NULL|  30|    PSG|
|      9|    Erling Haaland| 23| Norway|  28|    Man City|
|     10|    Harry Kane| 30| England|  24|Bayern Munich|
|     17|    Mohamed Salah| 32| Egypt|  21| Liverpool|
+-----+-----+-----+-----+-----+

```

```

from pyspark.sql.functions import avg, max, min

goals_summary = players_filled.agg(
    avg('Goals').alias('AverageGoals'),
    max('Goals').alias('MaximumGoals'),
    min('Goals').alias('MinimumGoals')
)

goals_summary.show()

```

```

+-----+-----+-----+
|AverageGoals|MaximumGoals|MinimumGoals|
+-----+-----+-----+
|      11.7|      30|      0|
+-----+-----+-----+

```